

Regularized Multivariate Two-way Functional Principal Component Analysis

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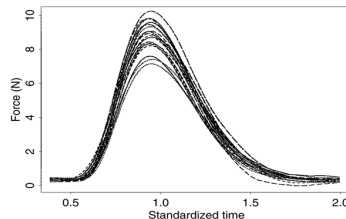
Outline

- 1 Introduction & Background
- 2 A SVD Approach for Regularized Multivariate FPCA
- 3 Two-way Regularized Multivariate FPCA
- 4 Conclusion & Future Work

Introduction

Functional Data Analysis:

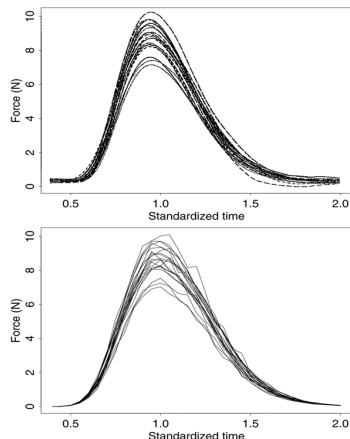
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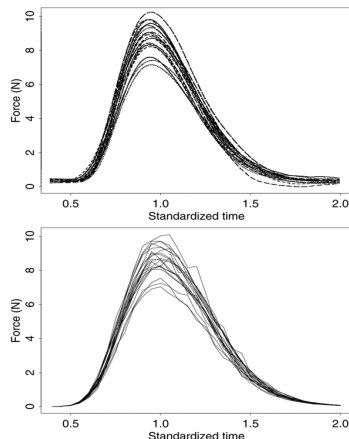
- **Functional data** are observations that change continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.
- In practice, these data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.



Introduction

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- In practice, these data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- **Functional Data Analysis (FDA)** is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.

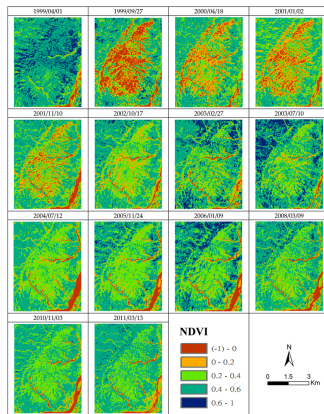


Functional Principal Component Analysis

- **Functional PCA (FPCA):** An extension of classical PCA for dimension reduction and uncovering hidden patterns in functional data; it identifies orthogonal functions that capture the main sources of variation, preserving the most important information. [Ramsay and Silverman, 2005].
- **Extensions of FPCA:**
 - Smoothed FPCA: Adds roughness penalties for smoothness [Silverman, 1996, Huang et al., 2008].
 - Sparse FPCA: Enforces sparsity for interpretability [Shen and Huang, 2008, Nie and Cao, 2020].
 - Multivariate FPCA (MFPCA): Extends FPCA to multivariate functions [Silverman, 1996, Happ and Greven, 2018].
 - Regularized MFPCA: Penalties improve estimation & interpretability [Hagbin et al., 2023].
- **Impact:** More adaptable, robust, and applicable across diverse scientific and business problems.

Two- way Functional Principal Component Analysis

- **Two-way functional data:** Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.



Two- way Functional Principal Component Analysis

- **Two-way functional data:** Observations vary along two domains (e.g., time $\times$ space, time $\times$ frequency), with applications in climate, neuroscience, finance, public health, and marketing.
- **Extension of FPCA:** Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- **Practical challenges:**
 - Data observed on discrete grids (minutes, hours, days).
 - Issues: measurement noise, irregular sampling, missing data, loss of smoothness.
- **Proposed framework:**
 - Unified FPCA for two-way multivariate functional data.
 - Smoothness penalties preserve functional structure.
 - Sparsity penalties enhance interpretability.
 - Effective for dimension reduction in complex datasets.

Foundations of FPCA through Minimizing Reconstruction Error

- **Goal:** Identify functional directions that maximize variance (low-rank approximation of functional data). For functional data $X \in \mathbb{R}^{n \times m}$ contains the discretized functional observations (rows correspond to subjects, columns to grid points), $v \in \mathbb{R}^m$ represents the estimated principal component (function), and $u \in \mathbb{R}^n$ denotes the associated principal component scores.
- **Reconstruction problem:**

$$\min_{u,v} \|X - uv^\top\|_F^2 = \text{tr}\{(X - uv^\top)(X - uv^\top)^\top\},$$

- **Optimization steps:**

$$\text{Fix } v : u = \frac{Xv}{v^\top v} \quad \text{and} \quad \text{Fix } u : v = \frac{X^\top u}{u^\top u}$$

Extensions of FPCA via Regularization

- **Goal:** Balance **variance explanation**, **smoothness**, and **interpretability**.
- Reformulate FPCA as a **penalized low-rank approximation** problem:

$$\min_{u,v} \|X - uv^T\|_F^2 + \mathcal{P}(u, v)$$

- Two directions:
 - **Smooth FPCA:** adds roughness penalty on functions.
 - **Sparse FPCA:** adds sparsity penalty on loadings.
- **Algorithms:** Based on iterative **power method** and **thresholding** updates.

Smooth Functional PCA

- Problem setup [Huang et al., 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha u^T u v^T \Omega v$$

- $X \in \mathbb{R}^{n \times p}$: discretized functional data.
 - $u \in \mathbb{R}^n$: scores.
 - $v \in \mathbb{R}^p$: loading function.
 - Ω : roughness penalty matrix (e.g., integrated squared 2nd derivative).
 - α : tuning parameter
- A power algorithm is defined to compute the PCs while incorporating smoothness penalty.

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- A power algorithm is defined to compute the PCs while incorporating smoothness penalty.
 - Consider the SVD of X as $X = UDV^\top$, where U and V have orthonormal columns and D is diagonal with ordered singular values. In particular, for $X = u_d v_d^\top$, v is the first principal component and $u = u_d$ gives the associated scores. With these representations, the power algorithm (described below) converges quickly, typically in only a few iterations.

Power Algorithm

Algorithm: Penalized Power Iteration

- ➊ Initialize v .
 - ➋ Repeat until convergence:
 - ➊ $u \leftarrow Xv$
 - ➋ $v \leftarrow (I + \alpha\Omega)^{-1}X^\top u$
 - ➌ $v \leftarrow \frac{v}{\|v\|}$
 - ➌ Update $X \leftarrow X - \sigma uv^\top$ and proceed to next component.
- For notational convenience, we define $S(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{m \times m}$, which simplifies expressions involving regularization. The penalty matrix Ω is set up so that larger values of the quadratic form $v^\top \Omega v$ mean rougher functions. This means that it penalizes functions that change quickly between time points.

Tuning Smoothness Parameters

- To select the optimal tuning parameters α efficiently, one can use a traditional Cross-Validation (CV) criterion and a computationally efficient closed-form Generalized Cross-Validation (GCV) criterion:

$$\text{CV}(\alpha) = \frac{1}{m} \sum_{j=1}^m \frac{\left[\{(I - S(\alpha))(X^T u)\}_{jj} \right]^2}{(1 - \{S(\alpha)\}_{jj})^2},$$

where $\{\cdot\}_{jj}$ denotes the j -th diagonal element.

$$\text{GCV}(\alpha) = \frac{1}{m} \frac{\|(I - S(\alpha))(X^T u)\|^2}{\left(1 - \frac{1}{m} \text{tr}\{S(\alpha)\}\right)^2}.$$

Sparse Functional PCA

- *Standard FPCA loadings are dense, involving linear combinations of all grid points $\rightarrow$ hard to interpret.*
Sparsity highlights only the most relevant features, thereby enhancing interpretability.
- *Dense loadings capture noise $\rightarrow$ unstable components.*
Sparsity filters out uninformative variation, yielding more robust principal components, reducing dimensionality, and facilitating interpretation.
- *All grid points contribute equally $\rightarrow$ no feature selection.*
Sparsity acts as an inherent feature selector, directing attention to key time points, with most entries reduced to zero while only a few contribute meaningfully to the structure.
- Sparse FPCA formulation [Shen and Huang, 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + p_\gamma(v) \quad (1)$$

where $p_\gamma(v)$ is a sparsity-inducing penalty.

Sparsity penalties

- **Soft-thresholding (Lasso):**

$$p_{\gamma}^{\text{soft}}(|\theta|) = 2\gamma|\theta|, \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{soft}}(y) = \text{sign}(y)(|y| - \gamma)_+$$

- **Hard thresholding:**

$$p_{\gamma}^{\text{hard}}(|\theta|) = \gamma^2 I(|\theta| \neq 0), \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{hard}}(y) = I(|y| > \gamma) y$$

- **SCAD penalty:**

$$p_{\gamma}^{\text{SCAD}}(|\theta|) = \begin{cases} 2\gamma|\theta|, & |\theta| \leq \gamma, \\ \frac{\theta^2 - 2a\gamma|\theta| + \gamma^2}{a-1}, & \gamma < |\theta| \leq a\gamma, \\ \frac{(a+1)\gamma^2}{2}, & |\theta| > a\gamma, \end{cases} \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{SCAD}}(y) = \begin{cases} \text{sign}(y)(|y| - \gamma)_+, & |y| \leq 2\gamma, \\ \frac{(a-1)y - \text{sign}(y)a\gamma}{a-2}, & 2\gamma < |y| \leq a\gamma, \\ y, & |y| > a\gamma, \end{cases}$$

where $a = 3.7$ (Fan and Li [2001]).

sFPCA-rSVD Algorithm

To implement the sPCA-rSVD algorithm discussed above, we use the following iterative procedure to minimize the objective function defined in Equation (1).

Algorithm: sFPCA-rSVD

- ➊ **Initialization:** Compute the best rank-one approximation of X using singular value decomposition (SVD), where $X \approx suv^\top$, and set $u \leftarrow su$.
- ➋ **Iterate until convergence:**
 - ➊ **Update Left Singular Vector:** $u \leftarrow Xv$
 - ➋ **Update Right Singular Vector:** $v \leftarrow h_\gamma X^\top u$
 - ➌ **Normalize Right Singular Vector:** $v \leftarrow \frac{v}{\|v\|}$

Cross-Validation for Sparsity Selection

- **Sparsity parameter:** Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV Tuning Parameter Selection - Degree of sparsity

- 1 Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \dots, X^K$.
- 2 For each sparse tuning parameter $j \in \{0, 1, \dots, p\}$ (level of sparsity), do the following:
 - 1 For $k = 1, \dots, K$, let X^{-k} be the data matrix X leaving out X^k . Apply Algorithm sFPCA-rSVD on X^{-k} and derive the FPC scores $u^{-k}(j)$. Then project X^k onto $u^{-k}(j)$ to obtain $v^k(j)$.
 - 2 Calculate the K-fold CV scores defined as: (N is the number of grid points in X^k)

$$CV_j = \sum_{k=1}^K \frac{\|X^k - u^{-k}(j)v^k(j)\|^2}{N}$$

- 3 Select the degree of sparsity as $j_0 = \arg \min \{CV(j)\}$.

Overview of Existing Approaches

- **Smooth FPCA:**

- Pros: Produces smooth eigenfunctions.
- Algorithm: Penalized power iteration.
- Tuning: α (smoothness) via CV/GCV.

- **Sparse FPCA:**

- Pros: Feature selection $\rightarrow$ interpretable.
- Algorithm: sFPCA-rSVD algorithm.
- Tuning: γ (sparsity) via CV.

- **Combined Approaches:** Smooth + Sparse together.

$$\min_{u,v} \|X - uv^T\|_F^2 + \alpha v^T \Omega v + p_\gamma(v)$$

- Trade-off: **Variance explained vs Interpretability vs Smoothness.**

Regularized MFPCA

- **Context**

- Univariate FDA → **Multivariate FDA** (e.g., simultaneously recorded EEG channels, growth patterns of multiple anatomical measures.)
- MFPCA → **joint modes of variation** across functions

- **Challenges**

- Discretization & irregular grids → noise, missing data
- High dimensionality and limited sample size → unstable eigenfunctions (sensitive to small fluctuations in the data)
- Cross-function correlation → requires enforcing smoothness both within and across functions

- **Proposed Solution: Penalized SVD**

- **Smoothness penalties:** roughness on derivatives
- **Sparsity penalties:** Soft, hard, or SCAD
- **Block-diagonal roughness matrix** for cross-function structure

- **Impact**

- Produces **smooth, sparse, interpretable** joint modes
- More stable & applicable to high-dimensional multivariate FDA

Methodology: Multivariate Functional Data Framework

- A multivariate functional dataset is formed by **concatenating** p **functional data matrices**.
 - Each variable: $X_i \in \mathbb{R}^{n \times m_i}$ where n : number of observations and m_i : grid points
- **Rank-one approximation** (per variable):

$$X_i \approx u_i v_i^\top, \quad u_i \in \mathbb{R}^n, \quad v_i \in \mathbb{R}^{m_i}$$

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- **Full data matrix:** $\mathbf{X} = [X_1 \quad X_2 \quad \cdots \quad X_p] \in \mathbb{R}^{n \times \sum_{i=1}^p m_i}$

$$\mathbf{X} = \begin{bmatrix} x_{11}(t_{11}) & \cdots & x_{11}(t_{1,m_1}) & \cdots & x_{1p}(t_{p1}) & \cdots & x_{1p}(t_{p,m_p}) \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1}(t_{11}) & \cdots & x_{n1}(t_{1,m_1}) & \cdots & x_{np}(t_{p1}) & \cdots & x_{np}(t_{p,m_p}) \end{bmatrix}.$$

Penalized Smooth MFPCA

- Standard FPCA loadings may be noisy; smoothness penalties (via block-diagonal Ω_i) improve structure and interpretability.
- Let $\mathbf{X} \in \mathbb{R}^{n \times M}$ denote multivariate functional data, where $M = \sum_{i=1}^p m_i$. Its best rank-one approximation is $\mathbf{X} \approx uv^\top$, with $u \in \mathbb{R}^n$ (score vector) and $v = [v_1, v_2, \dots, v_p]^\top \in \mathbb{R}^M$ (loading vector). A smoothness penalty is imposed on v .

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- The block-diagonal penalty matrix is $\mathbf{\Omega} = \text{diag}(\Omega_1, \Omega_2, \dots, \Omega_p)$, where each $\Omega_i \in \mathbb{R}^{m_i \times m_i}$ is a univariate roughness penalty matrix.

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- The penalized reconstruction error is

$$\min_{u,v} \|\mathbf{X} - uv^\top\|_F^2 + \alpha^\top (v^\top \mathbf{\Omega} v),$$

where $\alpha = (\alpha_1, \dots, \alpha_p)^\top$ controls smoothness.

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

- ➊ Initialize v .
 - ➋ Repeat until convergence:
 - ➊ $u \leftarrow \mathbf{X}v$
 - ➋ $v \leftarrow (I + \alpha\Omega)^{-1}\mathbf{X}^\top u$
 - ➌ $v \leftarrow v/\|v\|$
 - ➌ Update $\mathbf{X} \leftarrow \mathbf{X} - \sigma uv^\top$ to extract the next PC.
- The smoothing operator is $\mathcal{S}(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{M \times M}$.

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- The smoothing operator is $\mathbf{S}(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{M \times M}$.
 - The smoothing parameter α is selected via generalized cross-validation (GCV), defined as

$$\text{GCV}(\alpha) = \frac{1}{M} \frac{\|(I - \mathbf{S}(\alpha))(\mathbf{X}^\top u)\|^2}{\left(1 - \frac{1}{M}\text{tr}\{\mathbf{S}(\alpha)\}\right)^2}. \quad (2)$$

Penalized Sparse Multivariate FPCA

- **Goal:** Extend sparse FPCA to multivariate functional data, imposing **sparsity** (select important regions) and **smoothness** (reduce noise).
- **Sparsity penalties:** Soft, hard, or SCAD thresholding Shen and Huang [2008], Zhenhua Lin and Wang [2017], Nie and Cao [2020].

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- **Sparsity penalties:** Soft, hard, or SCAD thresholding Shen and Huang [2008], Zhenhua Lin and Wang [2017], Nie and Cao [2020].
- Sparsity parameters: $\gamma = (\gamma_1, \dots, \gamma_p)$, where γ_i ranges from 0 (no sparsity) to m_i for each variable.

Algorithm: Regularized Power Iteration for Smooth MFPCA

- 1 Initialization: Compute rank-one SVD of $\mathbf{X}$, $\mathbf{X} \approx \mathbf{su}\mathbf{v}^\top$, and set $\mathbf{u} \leftarrow \mathbf{su}$.
- 2 Iterate until convergence:
 - 1 Update left singular vector: $\mathbf{u} \leftarrow \mathbf{X}\mathbf{v}$
 - 2 Update right singular vector: $\mathbf{v} \leftarrow \mathbf{h}_\gamma \mathbf{X}^\top \mathbf{u}$
 - 3 Normalize right singular vector: $\mathbf{v} \leftarrow \frac{\mathbf{v}}{\|\mathbf{v}\|}$

Smooth and Sparse Multivariate FPCA

- The combined implementation of smoothness and sparsity on the loading vector v in multivariate functional data is achieved by the following algorithm:

Algorithm: Regularized Power Iteration for Smooth MFPCA

- Initialize unit vectors u and v using SVD of $\mathbf{X}$ (best rank-one approximation of $\mathbf{X}$)
 - Repeat till convergence
 - $u \leftarrow \mathbf{X}v$
 - $v \leftarrow \mathbf{S}(\alpha)\mathbf{h}(\gamma_v)\mathbf{X}^\top u$
 - $v \leftarrow \frac{v}{\|v\|}$
 - Update $\mathbf{X} = \mathbf{X} - \sigma uv^\top$ and proceed to find the next principal component.
- Algorithm CV Tuning for Sparsity and equation (2) are used to tune the sparsity level via K-fold CV and the smoothing parameter via GCV, respectively.

Simulation: Estimation Performance

- **Data-generating process:** Two functional variables:

$$X_{ij}^{(1)} = u_{i1}v_{11}(t_j) + u_{i2}v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1}v_{21}(t_j) + u_{i2}v_{22}(t_j) + \epsilon_{ij}^{(2)},$$

- where $u_{i1} \sim N(0, \sigma_1^2)$, $u_{i2} \sim N(0, \sigma_2^2)$, $\epsilon_{ij}^{(k)} \sim N(0, \sigma^2)$, and $n = m = 101$, $t_j \in [-1, 1]$

- **True functional PCs:**

- Variable 1: $v_{11}(t) = \frac{t + \sin(\pi t)}{s_1},$

$$v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$$

- Variable 2:

$$v_{21}(t) = \begin{cases} \frac{\sin(3\pi t)}{s_3}, & t \in (-\frac{1}{3}, \frac{1}{3}), \\ 0, & \text{otherwise,} \end{cases} \quad v_{22}(t) = \begin{cases} \frac{\sin(2\pi t)}{s_4}, & t \leq -\frac{1}{3}, \\ \frac{\sin(\pi t)}{s_4}, & t \geq \frac{1}{3}, \\ 0, & \text{otherwise.} \end{cases}$$

Here, s_1, s_2, s_3, s_4 are normalizing constants ensuring unit L^2 norm.

Simulation: Estimation Performance

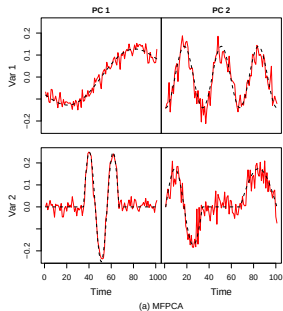
Scenarios tested:

1. Unpenalized Multivariate SVD (baseline)
2. Smoothed Multivariate SVD (smoothness penalty)
3. Sparse Multivariate SVD (sparsity penalty)
4. Sparse + Smoothed Multivariate SVD (combined regularization)

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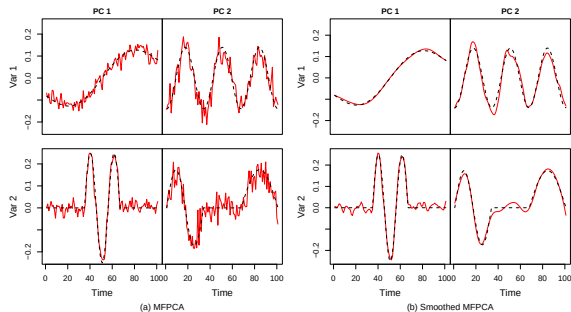
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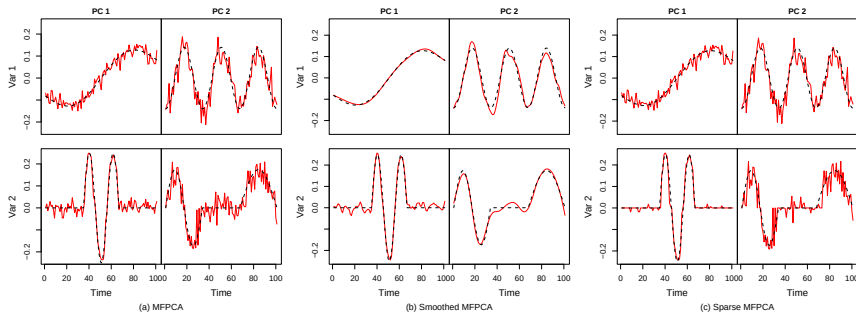
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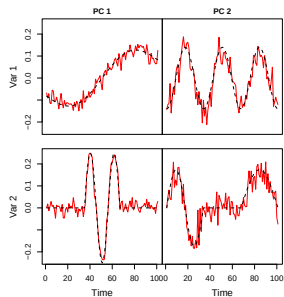
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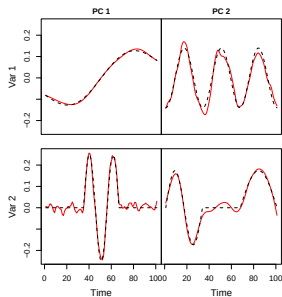
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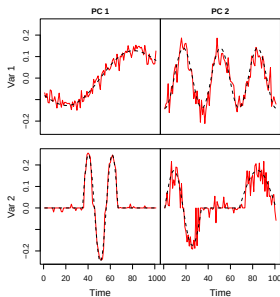
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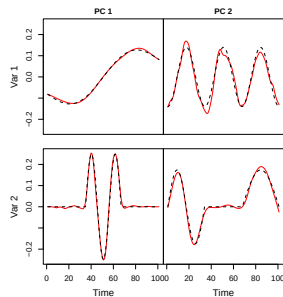
(a) MFPCA



(b) Smoothed MFPCA



(c) Sparse MFPCA



(d) Smoothed and Sparse MFPCA

Simulation: Estimation Performance

Accuracy measures:

- ① Variable-wise MSE:

$$\text{MSE}_{k\ell} = \frac{1}{m} \sum_{j=1}^m (\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

- ② Replication-averaged MSE:

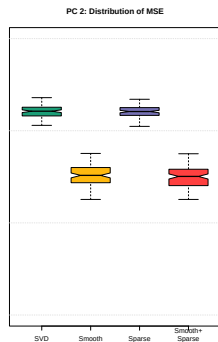
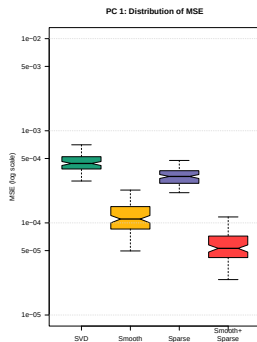
$$\overline{\text{MSE}}_{k\ell} = \frac{1}{R} \sum_{r=1}^R \text{MSE}_{k\ell}^{(r)}$$

- ③ Multivariate MSE:

$$\text{MSE}_{\ell}^{(\text{multi})} = \frac{1}{m} \sum_{j=1}^m \sum_{k=1}^2 (\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

Simulation: Estimation Performance

- **Performance across four methods (SVD, Smooth, Sparse, Smooth+Sparse):**
 - Smoothness and/or sparsity **reduce MSE** compared to unregularized SVD.
 - **Smooth+Sparse** yields lowest error and most stable estimates.
 - Smooth estimator performs consistently well; sparsity alone less effective (esp. for PC2).
 - Joint regularization achieves best **bias–variance tradeoff**.



PC1: Quartiles and Mean log₁₀(MSE)

Method	Q1	Median	Mean	Q3
SVD	-3.41	-3.35	-3.34	-3.28
Smooth	-4.07	-3.96	-3.92	-3.82
Sparse	-3.57	-3.50	-3.49	-3.43
Smooth+Sparse	-4.38	-4.28	-4.22	-4.14

PC2: Quartiles and Mean log₁₀(MSE)

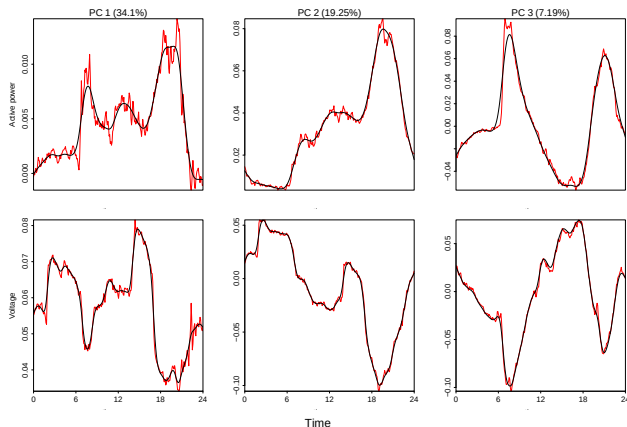
Method	Q1	Median	Mean	Q3
SVD	-2.84	-2.79	-2.79	-2.75
Smooth	-3.56	-3.48	-3.47	-3.40
Sparse	-2.83	-2.79	-2.79	-2.75
Smooth+Sparse	-3.59	-3.50	-3.49	-3.42

Application: Household Power Consumption

- **Dataset:** Bivariate functional data including active power and voltage consumption [Hebrail and Berard, 2012] for one household between December 2006 and November 2010.
- **Scaling:** To equalize the contribution of each variable in the multivariate analysis, we rescale them following [Happ and Greven, 2018].

$$\tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i),$$

$$\hat{w}_j = \left(\frac{1}{m} \sum_{i=1}^m \widehat{\text{Var}}(X_j(t_i)) \right)^{-1}.$$



First 3 PCs: MFPCA (red) vs ReMFPCA (black)

Regularization reduces noise while preserving the dominant daily consumption patterns, enhancing interpretability without losing key structure.

Two-way Regularized MFPCA

- **Two-way functional data:** Each observation is a *matrix of curves*, with smooth variation across **two domains** (e.g., time $\times$ space in air quality, time $\times$ channels in EEG).

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- **Our contribution:**
 - Extend to **multivariate functional data** (multiple functional variables).
 - Combine **smoothness + sparsity penalties** in both directions.
 - Result: Low-rank, interpretable, noise-robust principal components for high-dimensional applications.

Two-way Smoothed MFPCA: Setup & Penalty

- Two-way multivariate functional data: $\mathbf{X} \in \mathbb{R}^{n \times M}$, $M = \sum_{i=1}^p m_i$.
- Roughness matrices: $\mathbf{\Omega}_u \in \mathbb{R}^{n \times n}$, $\mathbf{\Omega}_v \in \mathbb{R}^{M \times M}$ (symmetric, non-negative definite).
- Smoothers: $\mathbf{S}_u(\alpha_u) = (\mathbf{I} + \alpha_u \mathbf{\Omega}_u)^{-1}$, $\mathbf{S}_v(\alpha_v) = (\mathbf{I} + \alpha_v \mathbf{\Omega}_v)^{-1}$.

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- Penalized rank-one reconstruction:

$$\min_{u,v} \|\mathbf{X} - uv^\top\|_F^2 + \mathcal{P}(u, v)$$

- Penalty [Jianhua Z. Huang and Buja, 2009]:
 $\mathcal{P}(u, v; \alpha_u, \alpha_v) = u^\top (\alpha_u \mathbf{\Omega}_u) u \|v\|^2 + \|u\|^2 v^\top (\alpha_v \mathbf{\Omega}_v) v + u^\top (\alpha_u \mathbf{\Omega}_u) u v^\top (\alpha_v \mathbf{\Omega}_v) v.$
- Multivariate v : $\mathbf{\Omega}_v = \text{diag}(\Omega_1, \dots, \Omega_p)$.

Two-way Smoothed MFPCA: Conditional GCV

- Minimizers:

$$u = \frac{S_u(\alpha_u) X v}{v^\top (I + \alpha_v \Omega_v) v} = \frac{S_u(\alpha_u)}{1 + \alpha_v R_v(v)} \frac{X v}{\|v\|^2}, \quad v = \frac{S_v(\alpha_v) X^\top u}{u^\top (I + \alpha_u \Omega_u) u} = \frac{S_v(\alpha_v)}{1 + \alpha_u R_u(u)} \frac{X^\top u}{\|u\|^2}.$$

- Rayleigh quotients: $R_u(u) = \frac{u^\top \Omega_u u}{\|u\|^2}, \quad R_v(v) = \frac{v^\top \Omega_v v}{\|v\|^2}.$

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- Conditional GCV criteria:

$$\text{GCV}_u(\alpha_u; \alpha_v) = \frac{\frac{1}{n} \left\| \left(I - \frac{\mathbf{S}_u(\alpha_u)}{1 + \alpha_v \mathbf{R}_v(v)} \right) \frac{\mathbf{X}_v}{\|v\|^2} \right\|^2}{\left(1 - \frac{1}{n} \text{tr} \left(\frac{\mathbf{S}_u(\alpha_u)}{1 + \alpha_v \mathbf{R}_v(v)} \right) \right)^2}, \quad \text{GCV}_v(\alpha_v; \alpha_u) = \frac{\frac{1}{m} \left\| \left(I - \frac{\mathbf{S}_v(\alpha_v)}{1 + \alpha_u \mathbf{R}_u(u)} \right) \frac{\mathbf{X}_u^\top}{\|u\|^2} \right\|^2}{\left(1 - \frac{1}{m} \text{tr} \left(\frac{\mathbf{S}_v(\alpha_v)}{1 + \alpha_u \mathbf{R}_u(u)} \right) \right)^2}.$$

- Optimization:** Alternate updates of u and v using GCV until convergence $\rightarrow$ two-way regularized components.

Two-way Smooth + Sparse MFPCA

- **Goal:** Extract components that are **low-rank, smooth, and sparse**.
- **Why?**
 - Smoothness → coherent variation across subjects & functions.
 - Sparsity → highlights key observations & time regions.
- **Novelty:** First framework to combine **both** in two-way functional data.
- **Result:** Robust, interpretable, noise-resistant components for high-dimensional data.

Sequential Power Algorithm

- Data matrix $\mathbf{X}$: seek u, v solving:

$$\min_{u,v} \|\mathbf{X} - uv^\top\|_F^2 + \sum_j \mathcal{P}_j^{[\alpha]}(u, v)$$

- Penalties: smoothness, sparsity (soft/hard thresholding, SCAD, group lasso, etc.).

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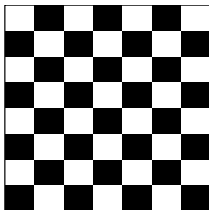
- Penalties: smoothness, sparsity (soft/hard thresholding, SCAD, group lasso, etc.).

Algorithm: Two-way Smooth + Sparse MFPCA (Sequential Power)

- 1 Initialization: Rank-one SVD of $\mathbf{X}$: $\mathbf{X} \approx s u^{(0)} v^{(0)\top}$; set $u \leftarrow s u^{(0)}$, $v \leftarrow v^{(0)}$.
- 2 Repeat until convergence:
 - 1 $u \leftarrow \mathbf{S}_u^{[\alpha_u]} \mathbf{h}_u^{[\gamma_u]}(\mathbf{X} v)$
 - 2 $v \leftarrow \mathbf{S}_v^{[\alpha_v]} \mathbf{h}_v^{[\gamma_v]}(\mathbf{X}^\top u)$
 - 3 $v \leftarrow v / \|v\|$
- 3 $\mathbf{X} \leftarrow \mathbf{X} - \sigma u v^\top$ to extract the next component.

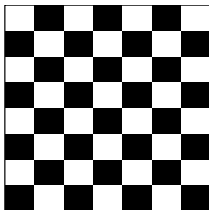
Chess Board

True Chessboard

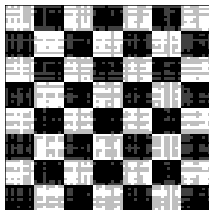


Chess Board

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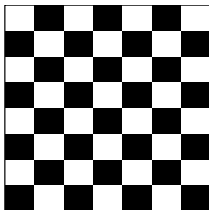


PCA with SVD

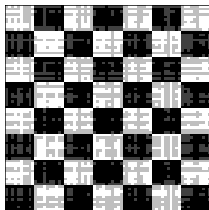


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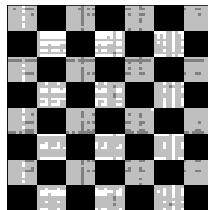
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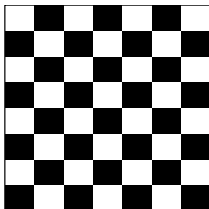


PCA with Group Lasso Penalty

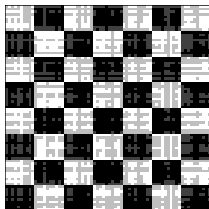


Chess Board

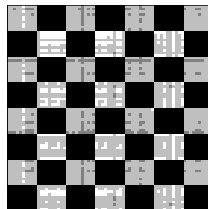
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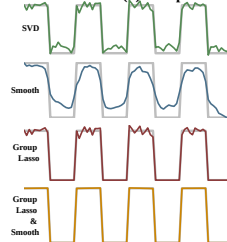
PCA with SVD



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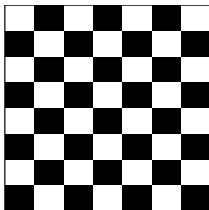


First PC Scores (u) Comparison

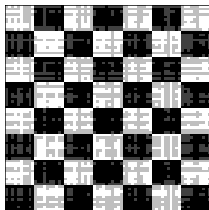


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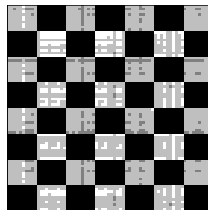
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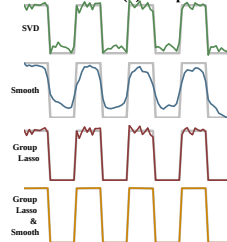
PCA with SVD



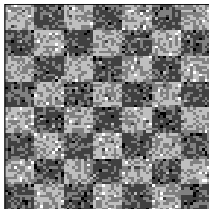
PCA with Group Lasso Penalty



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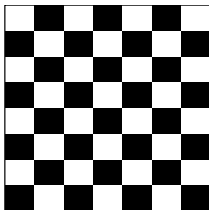


Noisy Chessboard

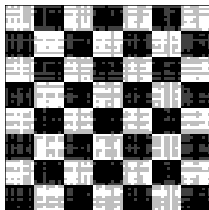


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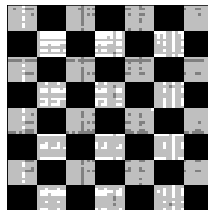
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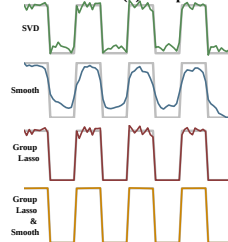
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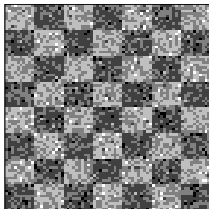
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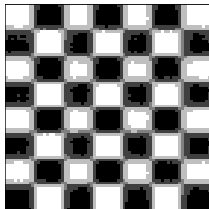
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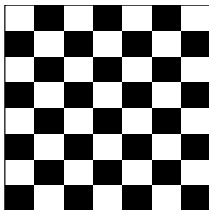


PCA with Smoothness Penalty

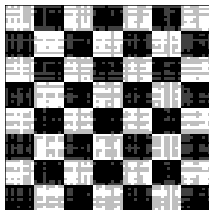


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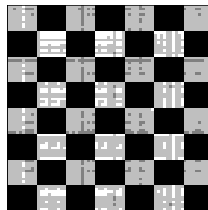
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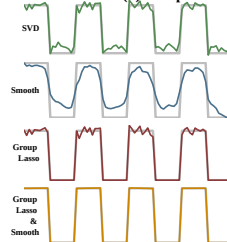
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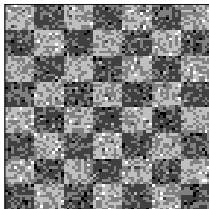
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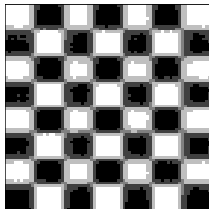
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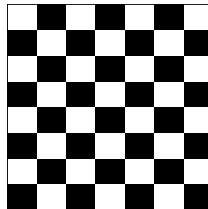
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PCA with Smoothness Penalty

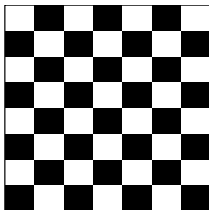


PCA with Group Lasso and Smoothness Penalties

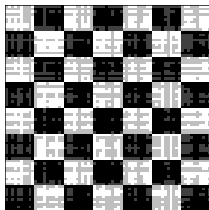


Chess Board

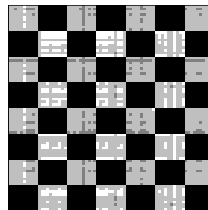
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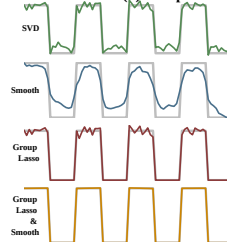
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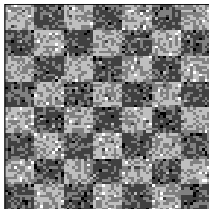
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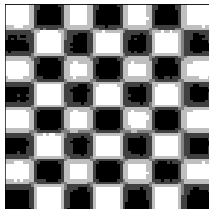
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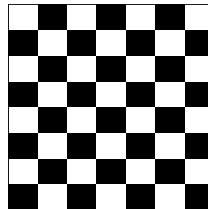
Noisy Chessboard



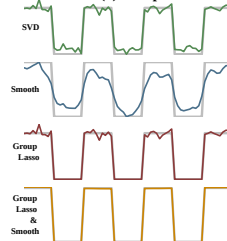
PCA with Smoothness Penalty



PCA with Group Lasso and Smoothness Penalties



First PC (v) Comparison



Selection of Regularization Parameters

- Four sets of tuning parameters:
 - α_u : smoothness of u , γ_u : sparsity of u
 - α_v : smoothness of v , γ_v : sparsity of v
- **Challenge:** Ordering of tuning (smoothness vs. sparsity) affects convergence and solutions.

Selection of Regularization Parameters

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 - α_u : smoothness of u , γ_u : sparsity of u
 - α_v : smoothness of v , γ_v : sparsity of v
- **Challenge:** Ordering of tuning (smoothness vs. sparsity) affects convergence and solutions.
- **Strategy: Conditional tuning**
 - 1 Initialize all penalties at 0.
 - 2 Tune γ_u via K -fold CV.
 - 3 Sequentially tune $\gamma_{v,i}$ using Algorithm: Two-way Smooth + Sparse MFPCA.
 - 4 With sparsity fixed, tune α_u by GCV.
 - 5 Tune $\alpha_{v,i}$ using two-way GCV.
 - 6 Iterate steps 2–5 until stable.
- This alternating scheme **isolates sparsity vs. smoothness** while ensuring accuracy + interpretability.

K-Fold CV algorithm for Sparsity

K-Fold CV (Row Sparsity)

- ❶ Split $\mathbf{X} \in \mathbb{R}^{n \times M}$ into K column groups $\{\mathbf{X}^{(1)}, \dots, \mathbf{X}^{(K)}\}$.
- ❷ For each γ_j and $k = 1, \dots, K$:
 - ❶ Train on $\mathbf{X}^{(-k)}$, estimate $u_j^{(-k)}$.
 - ❷ Validate: $v_j^{(k)} = \mathbf{X}^{(k)\top} u_j^{(-k)}$.
 - ❸ Fold error:

$$\text{Err}_j^{(k)} = \frac{1}{M} \|\mathbf{X}^{(k)} - u_j^{(-k)}(v_j^{(k)})^\top\|_F^2.$$

- ❸ CV score: $\widehat{\text{CV}}_j = \frac{1}{K} \sum_k \text{Err}_j^{(k)}$.
- ❹ Select $j_0 = \arg \min_j \widehat{\text{CV}}_j$.

K-Fold CV + 1-SE Rule

- ❶ Use same folds to collect $\text{Err}_j^{(k)}$.
- ❷ Compute mean $\widehat{\text{CV}}_j$ and SE $\widehat{\text{SE}}_j$:

$$\widehat{\text{SE}}_j = \sqrt{\frac{1}{K(K-1)} \sum_k (\text{Err}_j^{(k)} - \widehat{\text{CV}}_j)^2}.$$

- ❸ Let $j^* = \arg \min_j \widehat{\text{CV}}_j$.
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K-Fold CV algorithm for Sparsity

K-Fold CV (Column Sparsity)

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- ② For each j and $k = 1, \dots, K$:
 - ① Train on $\mathbf{X}^{(-k)}$, estimate $v_j^{(-k)}$.
 - ② Validate: $u_j^{(k)} = \mathbf{X}^{(k)} v_j^{(-k)}$.
 - ③ Fold error:

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Variance Explained: Classical vs Regularized FPCA

- **Classical FPCA:** Loadings v_j orthonormal; scores $u_j = Xv_j$ uncorrelated. Variance explained by first J PCs:

$$\sum_{j=1}^J \|u_j\|^2 = \text{trace}(V_J^\top X^\top X V_J), \quad V_J = [v_1, \dots, v_J].$$

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- **Issue under regularization:** smoothness/sparsity break orthogonality $\rightarrow$ scores become correlated $\rightarrow$ naive sum $\sum \|u_j\|^2$ **double-counts** variance (cf. Huang et al., 2008).

Subspace-Projection Definition of Explained Variance

- Normalize loadings and stack:

$$V_J = [v_1, \dots, v_J], \quad v_j \leftarrow \frac{v_j}{\|v_j\|}.$$

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- Orthogonal projector onto $\text{span}(v_1, \dots, v_J)$:

$$H_J = V_J (V_J^\top V_J)^{-1} V_J^\top,$$

where H_J is a symmetric idempotent matrix. ($H_J^2 = H_J$, $H_J^\top = H_J$)

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- Projected data and explained variance:

$$X_J = XH_J, \quad V_{\text{tot}} = \text{tr}(X^\top X), \quad \mathcal{V}_J = \|X_J\|_F^2 = \text{tr}(H_J X^\top X H_J).$$

PVE, Incremental PVE, and Properties

- Incremental variance:

$$\Delta \mathcal{V}_j = \mathcal{V}_j - \mathcal{V}_{j-1}, \quad \mathcal{V}_0 = 0.$$

PVE, Incremental PVE, and Properties

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- **Proportion of variance explained (PVE):**

$$\text{PVE}(J) = \frac{\mathcal{V}_J}{V_{\text{tot}}}, \quad \text{pve}_j = \frac{\Delta \mathcal{V}_j}{V_{\text{tot}}} = \text{PVE}(j) - \text{PVE}(j-1).$$

PVE, Incremental PVE, and Properties

- Incremental variance:

$$\Delta \mathcal{V}_j = \mathcal{V}_j - \mathcal{V}_{j-1}, \quad \mathcal{V}_0 = 0.$$

- **Proportion of variance explained (PVE):**

$$\text{PVE}(J) = \frac{\mathcal{V}_J}{V_{\text{tot}}}, \quad \text{pve}_j = \frac{\Delta \mathcal{V}_j}{V_{\text{tot}}} = \text{PVE}(j) - \text{PVE}(j-1).$$

- **Key properties:**

- No double-counting (works with correlated scores).
- Reduces to classical PCA when $V_J^\top V_J = I_J$.
- Monotone in J ($\mathcal{V}_J$ increases).
- $\Delta \mathcal{V}_j$ = unique variance added by component j .

Simulation: Two-way Functional Data

- Data-generating process:**

$$X_{ij}^{(1)} = u_{i1}v_{11}(t_j) + u_{i2}v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1}v_{21}(t_j) + u_{i2}v_{22}(t_j) + \epsilon_{ij}^{(2)},$$

- Latent scores:** generated as smooth functions

$$u_1(s) = \begin{cases} \sin(\pi s), & s > 0, \\ 0, & \text{otherwise,} \end{cases} \quad u_2(s) = \sin(2\pi s), \quad s \in [-1, 1].$$

- Functional PCs:**

- Variable 1: $v_{11}(t) = \frac{t + \sin(\pi t)}{s_1}, \quad v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$

- Variable 2:

$$v_{21}(t) = \begin{cases} \frac{\sin(3\pi t)}{s_3}, & t \in (-\frac{1}{3}, \frac{1}{3}), \\ 0, & \text{otherwise,} \end{cases} \quad v_{22}(t) = \begin{cases} \frac{\sin(2\pi t)}{s_4}, & t \leq -\frac{1}{3}, \\ \frac{\sin(\pi t)}{s_4}, & t \geq \frac{1}{3}, \\ 0, & \text{otherwise.} \end{cases}$$

Evaluation Metrics

- Integrated Squared Error (ISE):**

For replicate r and component u_1 :

$$\text{ISE}_r^{(u_1, \text{method})} = \frac{1}{m} \sum_{j=1}^m \left(u_1(t_j) - \hat{u}_1^{(\text{method})}(t_j) \right)^2.$$

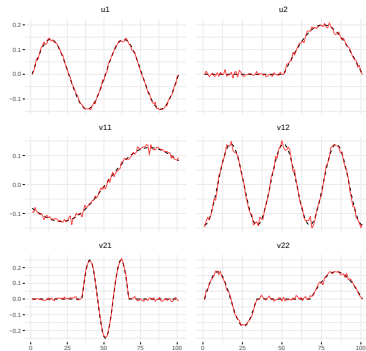
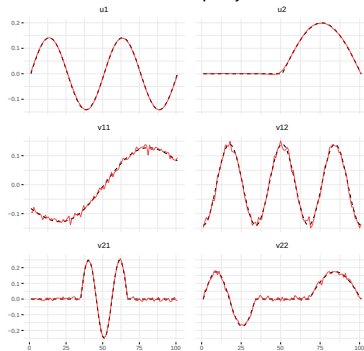
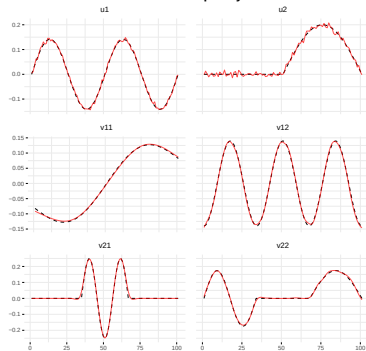
- Relative ISE (R_ISE):** ratio vs best method

$$R_r^{(u_1, \text{method})} = \frac{\text{ISE}_r^{(u_1, \text{method})}}{\text{ISE}_r^{(u_1, \text{best})}}.$$

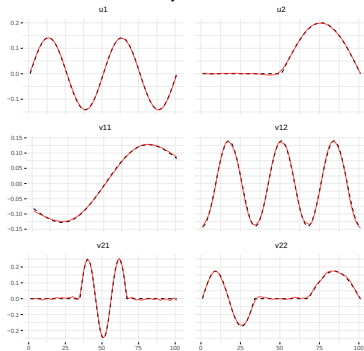
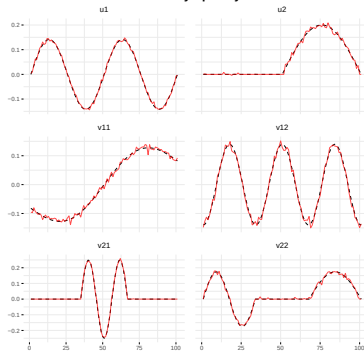
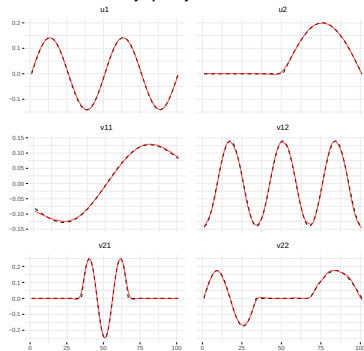
- Monte Carlo averages:**

$$\bar{R}^{(u_1, \text{method})} = \frac{1}{N} \sum_{r=1}^N R_r^{(u_1, \text{method})}, \quad \text{SE}(\bar{R}) = \sqrt{\frac{1}{N(N-1)} \sum_{r=1}^N \left(R_r - \bar{R} \right)^2}.$$

Simulation Results

SVD**Smoothness & Sparsity on u** **Smoothness & Sparsity on v** 

Simulation Results

Two-Way Smoothness**Two-Way Sparsity****Two-way Sparsity and Smoothness**

Simulation Results

Table 3: Mean ISE for each method and parameter

Method	u1	u2	v1	v2
SVD	0.3651	0.1587	0.00005	0.00010
Smooth+Sparse v	0.3651	0.1587	0.00002	0.00002
Smooth+Sparse u	0.3650	0.1584	0.00005	0.00010
Two-way Smoothness	0.3650	0.1585	0.00002	0.00002
Two-way Sparsity	0.3651	0.1586	0.00004	0.00009
Two-way Sm+Sp	0.3650	0.1584	0.00002	0.00002

Table 4: Mean Relative ISE for each method and parameter

Method	u1	u2	v1	v2
SVD	1.000	1.001	8.21	5.23
Two-way Sparsity	1.000	1.001	7.71	4.61
Smooth+Sparse v	1.000	1.001	1.05	1.01
Smooth+Sparse u	1.000	1.000	8.19	5.21
Two-way Smoothness	1.000	1.000	1.00	1.21

- Two-way Smooth+Sparse consistently yields the lowest errors across u and v .

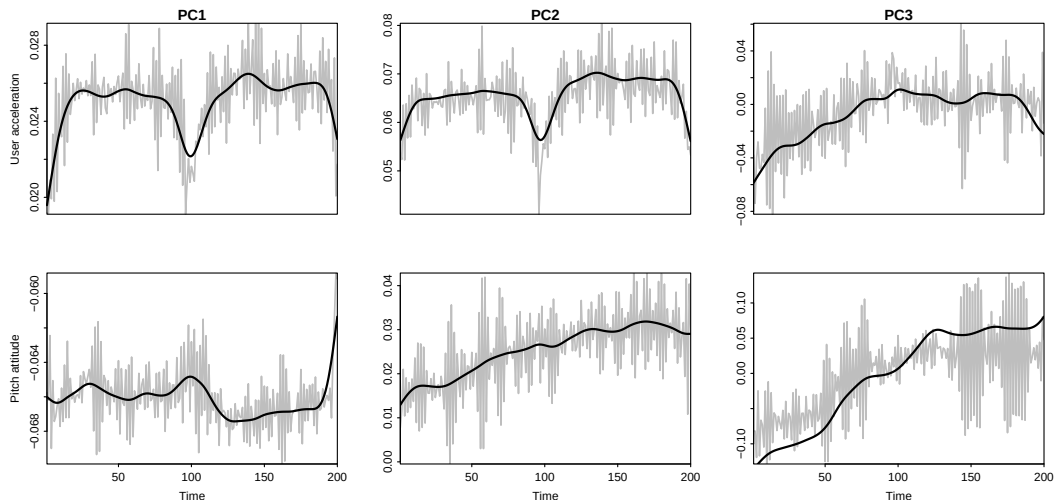
Application: Motion Sense Data

- **Dataset:** Acceleration and pitch from 24 people, 4 activities (jogging, walking, sitting, standing), about 2–3 min each.
- **Goal:** Compare SVD vs **two-way sparse + smooth ReMFPCA** on these multivariate functional signals.
- **Rescaling (Happ & Greven, 2018):** balance variables so each contributes equally.

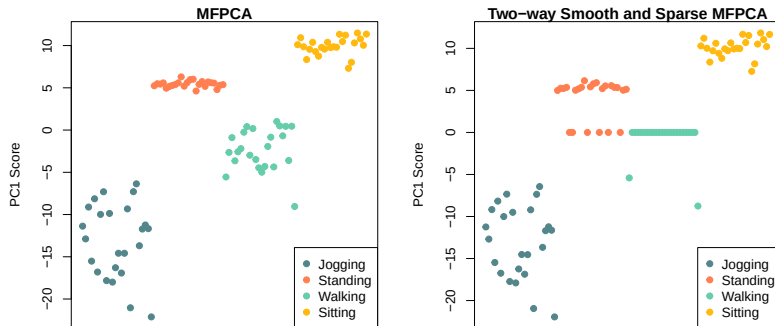
$$\hat{w}_j = \left(\frac{1}{m} \sum_{i=1}^m \widehat{\text{Var}}(X_j(t_i)) \right)^{-1}, \quad \tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i).$$

- **Penalties used:** smoothness + sparsity on loadings v ; sparsity on scores u (treated as random effects).

Results: Functional PCs (SVD vs ReMFPCA)



Results: PC Scores and Interpretation



- Sparsity on scores: **PC1 scores for walking about 0** (partially standing too).
- Interpretation: walking contributes little to PC1; removing it **improves interpretability** without hurting fit.
- Takeaway: **Two-way smooth + sparse ReMFPCA** yields cleaner PCs and activity-informative scores.

Conclusion & Future Work

- **Unified FPCA Framework**

- Combines **smoothness** (denoise, interpretability) + **sparsity** (variable selection).
- Extends from **univariate** → **multivariate** → **two-way functional data**.

- **Methodology**

- Penalized SVD with roughness + ℓ_1 penalties.
- Two-way regularization: smoothness & sparsity on both **scores** (u) and **loadings** (v).
- Efficient parameter tuning: **conditional GCV** & **K-fold CV** (with 1-SE rule).

- **Results**

- Simulations & applications (mortality, call-center, image data).
- Outperforms one-way or single-penalty methods.
- Produces **low-rank, denoised, interpretable components**.

Accessible Implementation: R Package & Future Work

- Implemented in **R package ReMPCA (GitHub)**

- Univariate & multivariate FPCA with penalties.
- Two-way MFPCA for matrix-valued functions.
- Automated tuning (CV, GCV, 1-SE rule).
- Diagnostic tools: variance explained, visualization, heatmaps.
- Early support for **hybrid data (scalar + functional + image)**.

- **Hybrid Data Extensions**

- Image–Functional Hybrid PCA → simultaneous dimension reduction.
- Scalar–Functional Integration → joint low-dim space.
- Nonlinear Extensions → kernel FPCA, neural nets.

- **Applications:**

- Neuroimaging
- Personalized medicine
- Environmental monitoring

Takeaway: Smooth + sparse + two-way FPCA offers a **theoretical foundation, practical algorithms, and open software** to enable next-generation functional data analysis.

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